

EXPERIMENT- 08

Python Code:

```
08.py > ...
1  alphabet = "abcdefghijklmnopqrstuvwxyz"
2  keyword = input("Enter keyword: ").lower()
3  plaintext = input("Enter plaintext: ").lower()
4
5  cipher_seq = ""
6  for char in keyword:
7      if char.isalpha() and char not in cipher_seq:
8          cipher_seq += char
9
10 for char in alphabet:
11     if char not in cipher_seq:
12         cipher_seq += char
13
14 print("Cipher Sequence:", cipher_seq.upper())
15
16 encrypted_text = ""
17 for char in plaintext:
18     if char.isalpha():
19         encrypted_text += cipher_seq[alphabet.index(char)].upper()
20     else:
21         encrypted_text += char
22
23 print("Encrypted Text:", encrypted_text)
24
25 decrypted_text = ""
26 for char in encrypted_text.lower():
27     if char.isalpha():
28         decrypted_text += alphabet[cipher_seq.index(char)]
29     else:
30         decrypted_text += char
31
32 print("Decrypted Text:", decrypted_text)
```

Output:

```
PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\08.py"
Enter keyword: CIPHER
Enter plaintext: abcdefghijklmnopqrstuvwxyz
Cipher Sequence: CIPHERABDFGJKLMNOQSTUVWXYZ
Encrypted Text: CIPHERABDFGJKLMNOQSTUVWXYZ
Decrypted Text: abcdefghijklmnopqrstuvwxyz
PS D:\College\Crypto\Experiments> |
```